

SOG XIAN, China

WMO: 561060

Lat: 31.8877N

Lon: 93.7796E

Elev: 13202

StdP: 8.91

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.5	1.4	-30.4	1.6	25.3	-25.9	2.1	23.7	25.1	35.8	22.6	33.7	1.1	290	0.686

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	70.2	49.8	67.6	49.4	65.2	48.7	53.7	63.1	52.2	62.4	51.0	61.1	5.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
50.8	91.6	57.0	48.7	84.4	55.5	47.2	79.8	54.8	28.6	62.9	27.4	62.7	26.4	61.4	61.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
21.3	17.3	14.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-9.1	75.2	4.9	2.4	-12.6	77.0	-15.4	78.4	-18.2	79.8	-21.7	81.5
			-9.9	55.2	4.2	2.0	-13.0	56.6	-15.5	57.8	-17.8	59.0	-20.9	60.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.9	19.0	23.6	30.2	37.7	44.6	51.4	55.1	54.5	49.4	39.0	28.5	21.0	
	DBStd	13.63	6.55	6.14	5.24	4.37	4.72	4.40	3.23	3.55	3.81	5.41	4.86	6.25	
	HDD50	4847	960	740	613	369	179	34	3	6	55	343	646	899	
	HDD65	9887	1425	1160	1078	819	631	407	306	326	468	807	1096	1364	
	CDD50	435	0	0	0	0	13	77	162	145	37	1	0	0	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH74	6	0	0	0	0	0	2	1	3	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	4.4	4.6	5.2	5.6	5.3	4.7	4.0	3.6	3.7	3.6	3.9	3.9	4.1	
Precipitation	PrecAvg	22.4	0.2	0.2	0.3	0.6	1.9	5.1	5.1	4.1	3.7	1.0	0.2	0.1	
	PrecMax	27.6	0.9	0.6	0.8	1.5	3.7	7.7	8.4	8.7	6.1	3.2	0.6	0.5	
	PrecMin	17.2	0.0	0.0	0.0	0.0	0.2	1.5	1.6	1.1	1.1	0.1	0.0	0.0	
	PrecStd	3.3	0.2	0.2	0.2	0.4	0.9	1.5	1.6	1.7	1.2	0.7	0.2	0.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.0	48.1	56.3	61.3	68.8	72.2	72.7	73.7	67.9	62.7	50.7	47.1	
		MCWB	29.5	28.9	32.4	35.3	43.4	48.9	51.7	49.5	48.7	41.0	30.2	28.0	
	2%	DB	42.0	43.6	50.1	56.3	63.3	67.2	69.7	70.3	64.5	57.2	46.9	42.2	
		MCWB	26.3	26.6	30.7	34.5	41.8	49.1	50.9	49.8	47.8	40.0	29.3	25.9	
	5%	DB	37.1	40.3	46.1	52.8	59.6	64.1	67.1	67.6	61.7	53.4	43.8	38.4	
		MCWB	23.3	24.9	28.9	33.4	40.1	48.1	50.0	49.9	46.8	38.2	28.1	24.0	
	10%	DB	32.8	36.9	42.9	49.8	56.1	61.1	64.4	64.6	58.9	49.9	40.6	34.7	
		MCWB	21.0	23.4	27.4	32.5	39.3	47.1	49.4	49.4	46.0	36.6	26.8	22.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.1	30.3	34.8	39.7	47.7	53.5	56.2	56.5	52.2	45.1	32.6	29.5
			MCDB	45.0	44.9	50.3	50.2	61.0	63.5	65.5	64.7	62.4	54.4	43.7	44.8
2%		WB	26.7	27.4	32.0	37.4	45.3	51.2	53.7	53.9	50.1	43.0	30.7	26.4	
		MCDB	40.0	40.7	46.5	48.6	57.5	62.2	63.9	62.1	59.9	53.4	42.4	40.0	
5%		WB	23.7	25.7	30.2	35.6	43.2	49.9	52.3	52.2	48.6	40.9	29.3	24.3	
		MCDB	35.3	38.0	43.2	48.1	54.1	60.8	62.9	61.7	58.2	49.4	40.8	36.7	
10%		WB	21.3	24.0	28.6	34.1	41.5	48.6	51.1	50.7	47.3	38.9	27.8	22.4	
		MCDB	31.4	35.3	40.3	46.3	51.2	58.3	61.4	60.7	56.5	46.4	38.8	33.6	
Mean Daily Temperature Range	5% DB	MDBR	25.3	23.4	22.2	21.5	19.4	17.8	18.6	19.6	19.5	19.7	24.3	26.8	
		MCDBR	33.2	30.0	28.4	28.3	26.6	22.8	23.5	25.8	24.2	26.0	30.2	33.1	
		MCWBR	20.5	17.8	14.5	12.5	10.1	8.5	8.5	8.8	9.6	10.9	15.7	20.4	
	5% WB	MCDBR	30.4	27.4	24.1	22.3	19.6	19.1	18.9	18.9	19.9	20.0	24.8	31.2	
		MCWBR	19.4	16.7	13.0	10.5	8.8	8.4	8.0	8.2	9.0	9.3	13.4	20.0	
Clear-Sky Solar Irradiance	taub	0.189	0.208	0.252	0.290	0.297	0.286	0.281	0.276	0.259	0.231	0.195	0.179		
	taud	2.648	2.578	2.454	2.387	2.447	2.583	2.654	2.654	2.672	2.674	2.699	2.738		
	Ebn at Noon	344	342	329	317	313	315	316	318	323	329	339	345		
	Edh at Noon	23	28	34	38	36	31	29	29	27	25	22	20		
All-Sky Solar Radiation	RadAvg	1125	1386	1684	1878	1937	1865	1859	1748	1562	1388	1276	1105		
	RadStd	108	80	110	115	123	112	116	126	80	90	59	75		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.06	N/A	N/A	N/A	-1.17	-1.26	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+0.92	+1.39	-3.51	N/A	N/A	N/A	-302	-392	+76	N/A

Nomenclature: See separate page